1)

a- support(d) = $\frac{5}{10}$ = 1/2

support(a,b) = $\frac{4}{10}$ = 2/5

support(b,d,e) = $\frac{1}{10}$

b- confidence($\{b,d\} \rightarrow \{a\}$) = 0/1 = 0

confidence($\{a\} \rightarrow \{b,d\}$) = 0/7 = 0

Confidence is not necessarily symmetric because although the output is 0, the denominator is different, so a different rule could yield two different outputs.

c- single-item

a	7
b	5
c	7
d	5
e	6

$L1 = \{a,b,c,d,e\}$

two-item

a,b	4
a,c	4
a,d	3
a,e	6
b,c	4
b,d	1

$L2 = \{(a,b), (a,c), (a,d), (a,e), (b,c), (b,e), (c,d), (c,e), (d,e)\}$

b,e	4
c,d	3
c,e	6
d,e	5

three-item

a,b,c	3
a,b,d	–
a,b,e	3
a,c,d	1
a,c,e	3
a,d,e	3
b,c,e	3
c,d,e	3

$L3 = \{(a,b,c),(a,b,e),(a,c,e),(a,d,e),(b,c,e),(c,d,e)\}$

$L4 = \{\}$

d- $a \to c : 6/7$

$b \to a : 4/5$

$b \to c : 4/5$

$b \to e : 4/5$

$c \to e = 6/7$

$d \to e : 5/5$

3 item sets

$\{a,b\} \to c : 3/4$

$\{a,c\} \to b : 3/4$

$\{b,c\} \to a : 3/4$

$\{a,b\} \to e : 3/4$

{b, e} → a = 3/4

{a, c} → e = 3/4

{a, d} → c = 3/3

{b, c} → e = 3/4

{b, e} → c : 3/4

{c, d} → e = 3/3

There is 16 association rules, 6 are from 2 item-sets and 10 are from 3 item-sets